

"Semiconductor"

(1) Semiconductor and its characteristics:-

Semiconductors are the materials whose electronic properties are intermediate b/w those of metals and insulators.

Semiconductors are used in many electrical circuits because we can control the flow of electrons in this material.

Silicon and Germanium are good semiconductor materials.

Characteristics :-

- (1) Their resistance decreases as their temperature increases, which is behaviour opposite to that of a metal.
- (2) Resistivity in range roughly 10^{-2} to 10^9 ohm cm.
- (3) Semiconductors have large thermoelectric power factor making them useful in thermoelectric generators.
- (4) Rectifying effects.
- (5) Sensitivity to light - either producing a photo voltage or change of conductivity.

Difference b/w intrinsic Semiconductor and Extrinsic Semiconductor:-

Intrinsic Semiconductor	Extrinsic Semiconductor
1. Pure form of Semiconductor	1. Impure form of Semiconductor
2. The Energy gap b/w conduction and valence band is small.	2. The Energy gap is higher than intrinsic Semiconductor.

3. Here Fermi level present in the middle of forbidden energy gap

3. The presence of Fermi level varies according to type of Extrinsic Semiconductor

4. It is not further classified.

4. It is classified as p-type and n-type Semiconductor

5. Ex: Silicon, Germanium etc.

5. Ex: GaAs, GeP etc.

Effect of temperature on Fermi level

(i) n-type Semiconductor → We know,

for n-type semiconductor number of electrons is more than number of holes that is mostly because there are additional electrons being supplied by donor atom (pentavalent atom).

So let us consider
 $n \approx N_D$

where N_D = concentration of Donor atoms.

as we know carrier concentration in n-type semiconductor is

$$n = N_c e^{-(E_c - E_f) / k_B T}$$

$$\text{So } N_D = N_c e^{-(E_c - E_f) / k_B T}$$

$$\text{Hence } N_c = 2 \left[\frac{2\pi m_e^* k_B T}{h^2} \right]^{3/2}$$

$$\frac{N_c}{N_D} = e^{(E_c - E_F)/k_B T}$$

$$\log \left(\frac{N_c}{N_D} \right) = \frac{E_c - E_F}{k_B T}$$

(Taking log both sides)

$$k_B T \log \left(\frac{N_c}{N_D} \right) = E_c - E_F$$

$$E_F = E_c - k_B T \log \left(\frac{N_c}{N_D} \right)$$

⇒ This equation shows that with rise in temp, fermi level drops.

This equation shows that rise in temperature, fermi level drops because at higher temperature semiconductor will produce more electron hole pair. Due to this production of extra e^- hole pair fermi level moves down toward the valence band.

(2) P-type Semiconductor

In P-type semiconductor concentration of holes can be approximately equal to $p \approx N_A$

Here N_A = concentration of acceptor atom (trivalent atom)

We know, for carrier concentration in P-type Semiconductor.

$$p = N_v e^{-(E_F - E_v)/k_B T} \quad \dots (1)$$

$$\text{Here } N_v = 2 \left[\frac{2 \pi m_p^* k_B T}{h^2} \right]^{3/2}$$

called density of state in valence band

So Equation (1) becomes,

$$n_p = n_v e^{-(E_F - E_v) / k_B T}$$

$$\frac{n_v}{n_p} = e^{(E_F - E_v) / k_B T}$$

$$\log\left(\frac{n_v}{n_p}\right) = \frac{(E_F - E_v)}{k_B T}$$

$$k_B T \log\left(\frac{n_v}{n_p}\right) = E_F - E_v$$

$$E_F = E_v + k_B T \log\left(\frac{n_v}{n_p}\right)$$

This equation shows that with rise in temperature fermi level rise toward conduction band. Because more temperature means more energy and electron tends to move towards the conduction band, which push the fermi level up.

Density or No. of holes in Valence band

We know that,

Holes means vacancy of electron
 $f(E)$ means occupational probability of electrons.

So, for holes probability function becomes,

$$1 - f(E) = 1 - \frac{1}{1 + e^{(E - E_F) / k_B T}}$$

$$= 1 - \left[1 + e^{(E - E_F) / k_B T} \right]^{-1}$$

$$= 1 - \left[1 - e^{(E - E_F) / k_B T} \right]$$

$$1 - f(E) = e^{(E - E_F) / k_B T} \quad \dots (1)$$

for top of valence band (maximum energy)

density of states is

$$N(E) = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (E_v - E)^{1/2} \quad \text{--- (2)}$$

Here m_p^* is effective mass of hole near top of valence band.

Now, number of Holes is

$$P = \int_{-\infty}^{E_v} N(E) [1 - f(E)] dE \quad \text{--- (3)}$$

Now put the value of $N(E)$ and $[1 - f(E)]$ from Equation (1) and (2) in eq (3)

Conduction band	$+\infty$
	E_c
Forbidden Gap	E_v
valence band	$-\infty$

$$P = \int_{-\infty}^{E_v} \frac{4\pi}{h^3} (2m_p^*)^{3/2} (E_v - E)^{1/2} e^{(E - E_f)/k_B T} dE$$

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} \int_{-\infty}^{E_v} (E_v - E)^{1/2} e^{(E - E_f)/k_B T} dE$$

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (k_B T)^{1/2} \int_{-\infty}^{E_v} \left(\frac{E_v - E}{k_B T} \right)^{1/2} e^{(E - E_f + E_v - E_v)/k_B T} dE$$

($\therefore$ dividing numerator and denominator by $(k_B T)^{1/2}$ and add and subtracting

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (k_B T)^{1/2} e^{(E_v - E_f)/k_B T} \int_{-\infty}^{E_v} \left(\frac{E_v - E}{k_B T} \right)^{1/2} e^{(E - E_v)/k_B T} dE \quad \text{--- (4)}$$

let us put $\frac{E_v - E}{k_B T} = x$

$$-dE/k_B T = dx$$

So, $dE = -(k_B T) dx$

Equation (4) becomes,

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (k_B T)^{1/2} e^{(E_v - E_F)/k_B T} \int_0^\infty x^{1/2} e^{-x} (-k_B T) dx$$

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (k_B T)^{1/2} e^{(E_v - E_F)/k_B T} \int_0^\infty x^{1/2} e^{-x} (k_B T) dx$$

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (k_B T)^{1/2} e^{(E_v - E_F)/k_B T} \int_0^\infty x^{1/2} e^{-x} dx$$

$$\text{as } \int_0^\infty x^{1/2} e^{-x} dx = \frac{\sqrt{\pi}}{2}$$

So above equation becomes,

$$P = \frac{4\pi}{h^3} (2m_p^*)^{3/2} (k_B T)^{1/2} e^{(E_v - E_F)/k_B T} (k_B T) \frac{\sqrt{\pi}}{2}$$

$$P = 2 \left[\frac{2\pi m_p^* k_B T}{h^2} \right]^{3/2} e^{(E_v - E_F)/k_B T}$$

This equation is concentration of Holes in valence band.

or $P = N_v e^{(E_v - E_F)/k_B T}$

$$\text{Here } N_v = 2 \left[\frac{2\pi m_p^* k_B T}{h^2} \right]^{3/2}$$

Density of ^{or} no. of Electrons in conduction Band

Let n_e → density of electrons per unit volume in conduction Band at temperature T .
Electron lies in conduction band from E_c to ∞ .

We know,

According to free electron theory, density of states per unit volume is given by,

$$D(E) = \frac{4\pi}{h^3} (2m)^{3/2} E^{1/2} \quad \text{--- (1)}$$

Now, Density of electrons in conduction Band is

$$n_e = \int_{E_c}^{\infty} D(E) f(E) dE \quad \text{--- (2)}$$

Here $f(E)$ = fermi-dirac distribution

E_c = Energy of bottom of conduction band

$$f(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1} \quad \text{--- (3)}$$

Replace E by $E - E_c$ in equation (1) and m by m^* in eq (1)

$$D(E) = \frac{4\pi}{h^3} (2m^*)^{3/2} (E - E_c)^{1/2} \quad \text{--- (4)}$$

Using equation (3) and (4) and eq (2) becomes,

$$n_e = \int_{E_c}^{\infty} \frac{4\pi}{h^3} (2m^*)^{3/2} (E - E_c)^{1/2} \times \frac{1}{e^{(E-E_F)/k_B T} + 1} dE$$

for $E - E_F \gg K$

$$e^{(E-E_F)/K_B T} + 1 \approx e^{(E-E_F)/K_B T}$$

$$n_e = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_c}^{\infty} \frac{(E-E_c)^{1/2} dE}{e^{(E-E_F)/K_B T}}$$

$$n_e = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_c}^{\infty} (E-E_c)^{1/2} \cdot e^{(E_F-E)/K_B T} dE$$

$$n_e = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_c}^{\infty} (E-E_c)^{1/2} \exp \left[\frac{E_F - E_c + E_c - E}{K_B T} \right] dE$$

$$n_e = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int (E-E_c)^{1/2} \exp \left(\frac{E_F - E_c}{K_B T} \right) \exp \left(\frac{E_c - E}{K_B T} \right) dE$$

-----(5)

Put $\frac{E-E_c}{K_B T} = x$

$$\Rightarrow (E-E_c) = x K_B T$$

$$\Rightarrow (E-E_c)^{1/2} = x^{1/2} (K_B T)^{1/2} \quad \dots \dots 5(a)$$

Also, $dE = K_B T dx \quad \dots \dots 5(b)$

Using 5(a) and 5(b), eq (5) becomes

$$n_e = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \exp \left[\frac{E_F - E_c}{K_B T} \right] \int_{E_c}^{\infty} x^{1/2} (K_B T)^{1/2} e^{-x} K_B T dx$$

-----(6)

Now, $n_e = \frac{4\pi}{h^3} (K_B T)^{3/2} (2m_e^*)^{3/2} \exp \left[\frac{E_F - E_c}{K_B T} \right]$

$$\int_{E_c}^{\infty} x^{1/2} e^{-x} dx \quad \dots \dots (7)$$

Using standard integral

$$\int_{E_c}^{\infty} e^{-x} \cdot x^{1/2} dx = \sqrt{\pi}/2$$

So equation (7) becomes,

$$n_c = \frac{4\pi}{h^3} (k_B T)^{3/2} (2m_e^*)^{3/2} \exp\left[\frac{E_F - E_c}{k_B T}\right] \cdot \frac{(\pi)^{1/2}}{2}$$

$$n_c = 2 \left[\frac{2m_e^* k_B T \pi}{h^2} \right]^{3/2} \exp\left[\frac{E_F - E_c}{k_B T}\right]$$

This is concentration (density) of electron in conduction Band in intrinsic Semiconductor.

$$n = N_c e^{-(E_c - E_F)/k_B T}$$

$$\text{Here } N_c = 2 \left[\frac{2\pi m_e^* k_B T}{h^2} \right]^{3/2}$$

Carrier Concentration in intrinsic Semiconductor:-

- (1) Density of electrons in conduction band.
- (2) Density of holes in valence band.
- (3) Fermi level.

Fermi level in intrinsic Semiconductor

In intrinsic Semiconductor,

$$\text{No. of electrons} = \text{no. of holes}$$
$$n = p$$

So,

$$N_C e^{-(E_C - E_F)/k_B T} = N_V e^{-(E_F - E_V)/k_B T}$$

Taking log both sides and rearranging terms-

$$-\left(\frac{E_C - E_F}{k_B T}\right) = \log\left(\frac{N_V}{N_C}\right) - \frac{(E_F - E_V)}{k_B T}$$

Now multiply both sides by $k_B T$.

$$-(E_C - E_F) = k_B T \log\left(\frac{N_V}{N_C}\right) - (E_F - E_V)$$

$$2E_F = k_B T \log\left(\frac{N_V}{N_C}\right) + (E_C + E_V)$$

$$E_F = \frac{E_C + E_V}{2} + \frac{1}{2} \log k_B T \log\left(\frac{N_V}{N_C}\right) \quad \dots (1)$$

We know,

$$N_V = 2 \left[\frac{2 \pi m_p^* k_B T}{h^2} \right]^{3/2}$$

$$N_C = 2 \left[\frac{2 \pi m_e^* k_B T}{h^2} \right]^{3/2}$$

Now put the values of N_V and N_C in equation (1) we get,

$$E_F = \frac{(E_C + E_V)}{2} + \frac{3}{2 \times 2} k_B T \log\left(\frac{m_p^*}{m_e^*}\right)$$

as intrinsic Semiconductor,

$$m_p^* = m_e^*$$

So above equations,

$$E_F = \frac{E_C + E_V}{2}$$

Thus, Fermi level in intrinsic Semiconductor lies exactly half way between top of valence band and bottom of conduction band i.e. centre of forbidden energy gap.

Fermi level in Extrinsic Semiconductor:

In Extrinsic Semiconductors, the number of electrons in conduction band and number of Holes in valence band are not equal. Hence, the probability of Occupation of energy level in conduction band and valence band are not equal. Therefore, the Fermi level for extrinsic Semiconductor lies close to conduction or valence band.

Fermi level in N-type Semiconductor:

In N-type Semiconductor pentavalent impurities are added. Each pentavalent impurity donates free e^- . The addition of pentavalent impurity creates large number of free electron in conduction band.

At room temp., the number of electrons in conduction band is greater than number of Holes in valence band. Hence, the probability of occupation of energy levels by electrons in conduction band is greater than the probability of occupation of energy levels by Holes in valence band. This probability of occupation of energy levels is represented in terms of Fermi level.

Therefore, the Fermi level in n-type Semiconductor lies close to the conduction band.

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Fermi level for N-type Semiconductor is

$$E_F = E_C - K_B T \ln \frac{N_C}{N_D}$$

Here E_F = Fermi level

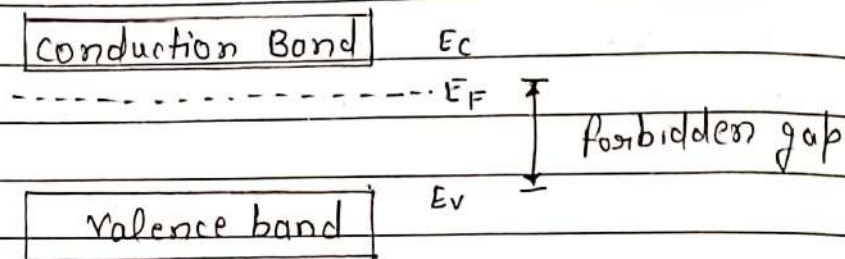
E_C = conduction band

K_B = Boltzmann constant

T = absolute Temp.

N_C = density of states in conduction band

N_D = concentration of donor atoms.



* Fermi level in p-type Semiconductor is

In p-type

Semiconductor trivalent impurity is added.

Each trivalent impurity creates a hole in valence band and ready to accept an electron.

The addition of Trivalent impurity creates large number of holes in valence band.

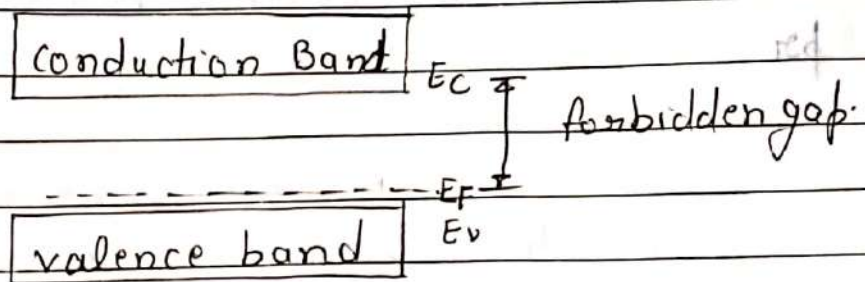
At room temp. number of holes in valence band is greater than the number of electrons in conduction band. Hence the probability of occupation of energy levels by holes in valence band is greater than probability of occupation of energy levels by electron in conduction band. This probability of occupation of energy level is represented

by fermi level.

Therefore, the fermi level in p-type Semiconductor lies close to valence band and is given as:

$$E_F = E_V + K_B T \log \frac{N_V}{N_A}$$

Here N_V = density of states in Valence band
 N_A = concentration of acceptor atoms.



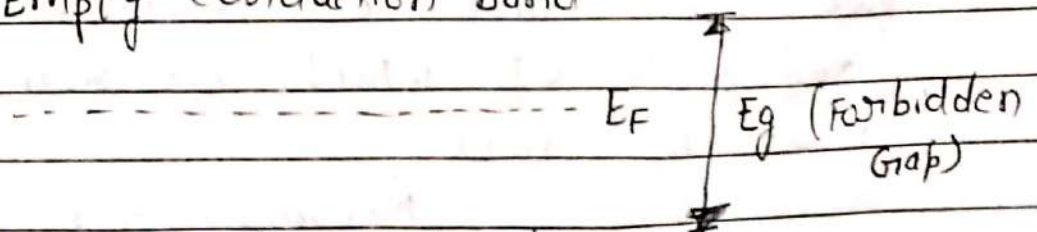
Carrier Generation and Recombination

Carrier Generation

and recombination may be defined as "Generation is process where electron and holes are created and recombination is the process where electron and holes are annihilated."

In Semiconductor, at absolute zero all the energy levels below fermi level are completely filled, while all the energy levels above fermi level are completely empty.

Empty Conduction Band



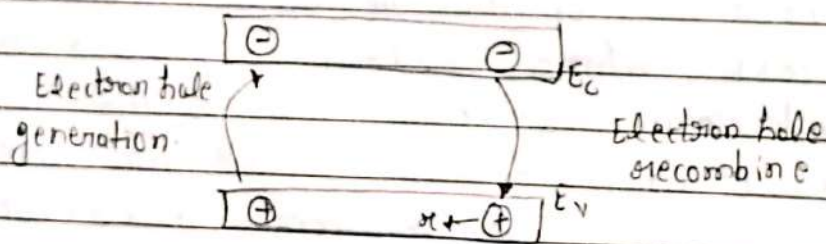
filled Valence band

In intrinsic semiconductor, Fermi level lies in the middle of valence band and conduction band.

If an electron in valence band acquire energy greater than forbidden gap energy. Then, electron will reach into conduction band and leave a hole in valence band.

Now, this electron will freely move in the conduction band. Thus, in carrier generation process two mobile charge carriers (electron and holes) are generated.

At the same time, electrons moving in conduction band may come in close proximity to holes and fall in the empty states in valence band. This recombination process annihilates both electrons and holes as shown in fig.



Net charge carrier concentration are independent of time in thermal equilibrium so the rate of electron hole pair generated and rate of which electron and holes recombine must be equal.

Recombination and Generation are always happening in semiconductor and their rates are constant at equilibrium. so.

product of electron and hole densities is constant
i.e.

$$np = n_i^2$$

At equilibrium.

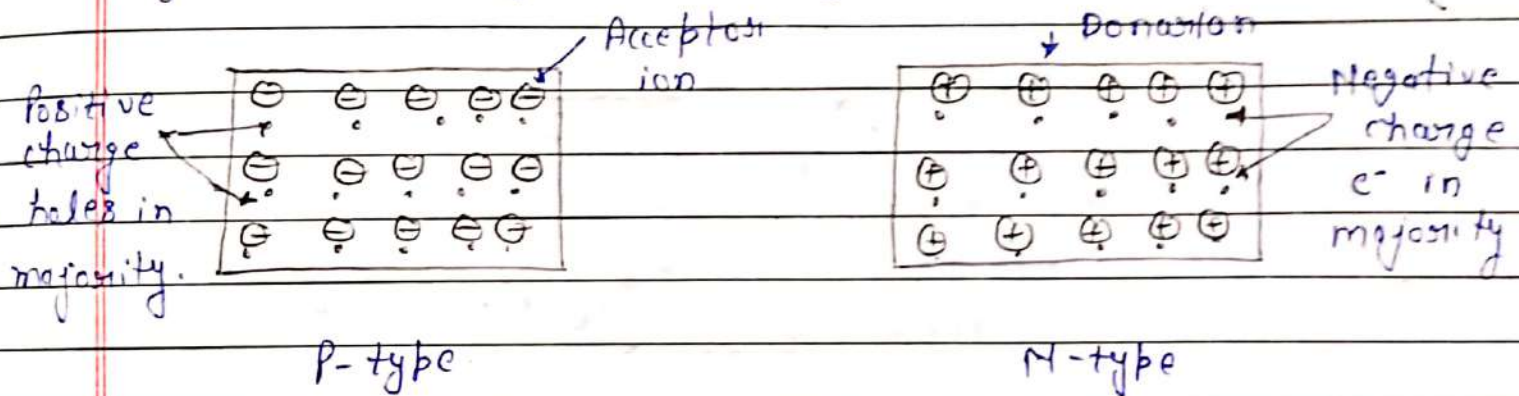
Carrier Generation and Recombination process is fundamental to operation of many optoelectronic semiconductor devices such as photodiodes, LED and laser diodes.

"P-N Junction"

P-N Junction is formed by diffusing P-type material to one half side and N-type material to other half side. The plane dividing the two zones is known as Junction.

P-type and N-type semiconductor pieces before joining is shown in fig.

In fig. P-type semiconductor has negative acceptor ion (shown by encircled minus sign) and positive charge holes. Similarly in N-type semiconductor has positive donor ions (shown by positive sign) and negative charged free electrons.



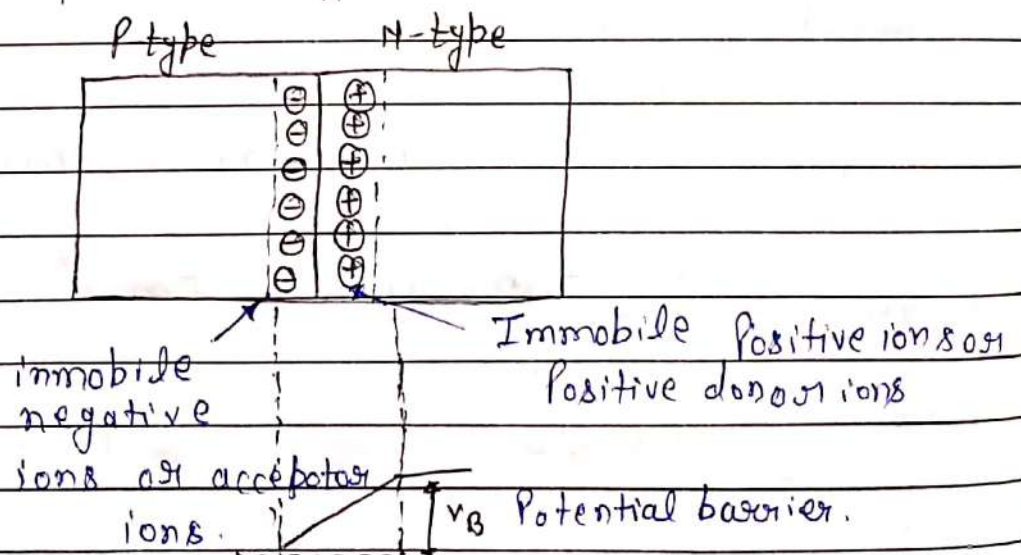
P-type and N-type Semiconductor before joining.

Formation of depletion layer :-

Now, consider that two pieces are joined together as shown in fig. below. As P-type material has high concentration of holes and N-type material has high concentration of electrons. Hence there is tendency of holes to diffuse over N-side and electrons to P-side. This process is called "Diffusion".

So due to diffusion some of holes from p-side cross over to N-side and become neutral. Similarly, some of electrons from N-side cross over to p-side where they combine with holes and become neutral.

Thus a region is formed known as "depletion region" or space charge region. The diffusion of holes and electrons continues till potential barrier in space charge region which prevents further diffusion or neutralisation.



Some points :-

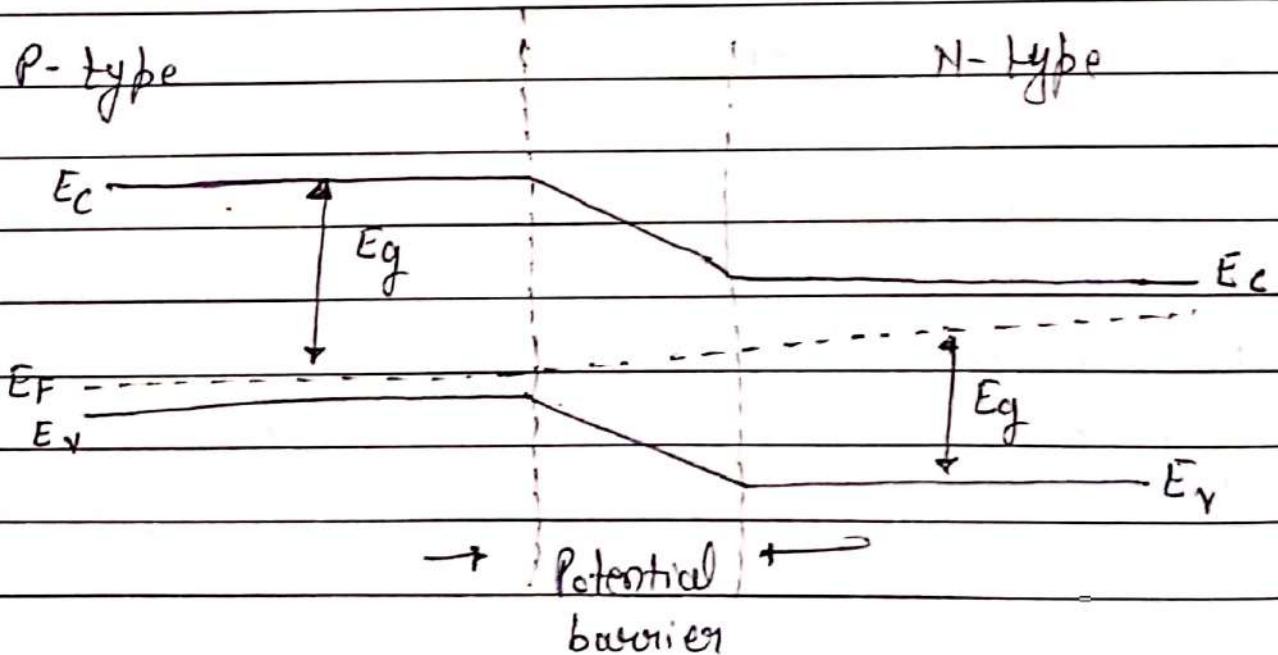
1. When P-N Junction is formed, the holes from P-region diffuse to N-region where they dis-

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- appear due to recombinations with electrons.

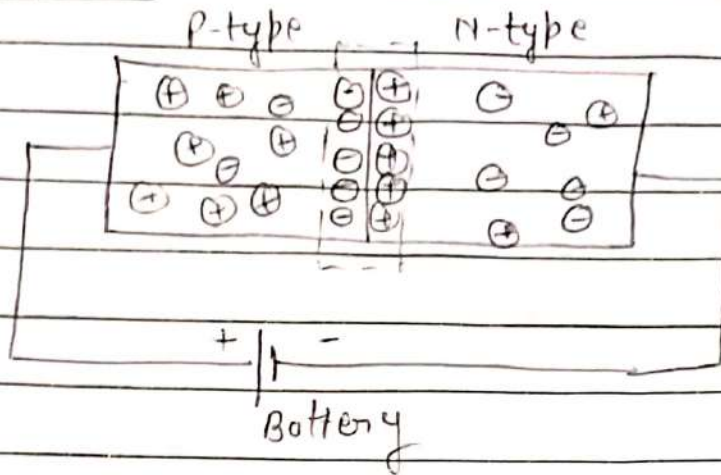
2. The electrons from N-region diffuse to P-region where they disappear due to recombination with holes.
3. In this procedure, negative immobile acceptor ions in P-region and positive immobile donor ions in N-region are left near Junction.
4. Now, no further diffusion of holes and electrons take place across the Junction because holes trying to diffuse in N-region are repelled by immobile positive ions and the electrons trying to diffuse in P-region are repelled by immobile negative ions.
5. The region containing immobile negative and immobile positive ions is called "Depletion Region."
6. As negative ions (immobile) are created on P-side of Junction so P-type acquires negative potential. Similarly positive ions are created on N-side so N-side acquires positive potential. Hence the voltage ' V_B ' is developed across the Junction under equilibrium condition. This voltage is known as Junction Voltage or internal voltage or potential barrier voltage ' V_B ' which is about 0.3 V for Ge and 0.7 V for Si at room temp 300K.
7. We know Fermi level lies near the valence band in p-type and near the conduction band in n-type semiconductors. Under thermal equilibrium, the average energy of P and N region becomes equal.

So Fermi level (E_F) of P and N region will be at same level as shown in fig. below. The equilibrium of fermi level in P and N type is analogous to equilibrium of level of water in two containers on ~~being~~ being joined together.



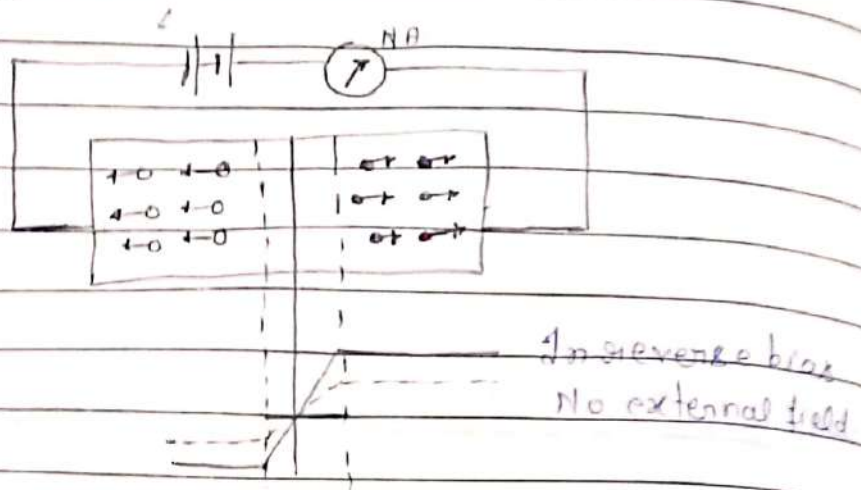
Working OF P-N Junction Diode :-

ii) Forward Bias :-



In the forward bias, the negative terminal of battery is connected to N-type semiconductor and positive terminal is to P-type materials. This connection is called also as giving positive voltage. Electrons from the N-region crosses the Junction and enter the P-region. Due to the attractive force that is generated in the P-region the electrons are attracted and move towards positive terminal. Similarly the holes are attracted towards negative terminal. By the movement of Electrons and holes current flows. In this condition the width of depletion layer decrease due to reduction in number of positive and negative ions.

Reverse bias :-



In reverse bias, the negative terminal is connected to P-type and positive terminal of battery is connected to n-type materials. Hence, the electric field due to both the voltage and depletion layer is in the same direction. This makes the electric field more stronger than before. Due to this strong electric field, electrons and holes want more energy to cross the junction so they cannot diffuse to opposite region. Hence, no current flow due to lack of movement of electrons and holes. The electrons from the N-type semiconductor are attracted towards the positive terminal and the holes from the P-type semiconductor are attracted towards negative terminal. This leads to the reduction of the number of electrons in N-type and holes in P-type. In addition, positive ions are created in N-type region and negative ions are created in P-type region.

Therefore the depletion layer width is increased due to the increasing number of positive and negative ions.

Difference b/w $\circ +$

Ideal diode

1. Ideal diode act as a perfect conductor and perfect insulator.
2. Ideal diode draw no current when reverse biased.
3. Ideal diode offers infinite resistance when reverse biased.
4. It ~~cannot~~ cannot be manufactured.
5. It has zero cut-in voltage.
6. Ideal diode has zero voltage drop across its junction when forward bias.

Practical or Real diode.

1. Practical diodes cannot act as perfect conductor and perfect insulator.
2. This diode draws very low current when reverse biased.
3. Practical diode offers very high resistance when reverse biased.
4. It can be manufactured.
5. It has very low cut in voltage.
6. It has very low voltage drop across it when forward bias.

As the forward bias increase the conductivity of

Metal - Semiconductor Junctions.

The interface between semiconductor materials and insulator is known as Junction. So three types of Junctions can be formed.

1. Metal-metal Junction (used to make thermocouples)
2. Metal - Semiconductor Junction (Schottky barrier or ohmic contact)
3. Semiconductor - Semiconductor Junction (P-n Junction)

Metal to semiconductor junctions are of great importance as it affects the electrical properties of the devices. They can behave either as a Schottky barrier or ohmic barrier contact depending upon the characteristics of interfaces.

Schottky Junctions:-

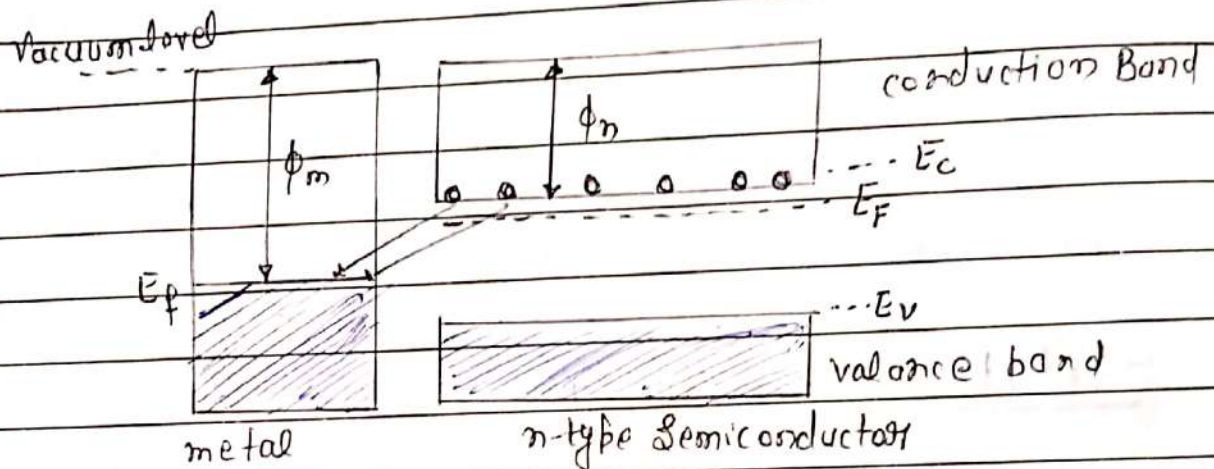
Let us consider the junction formed between metal and n-type semiconductor. Since work function of semiconductor is lower than that of metals so its Fermi level will be higher than that of metal. The Fermi level will be line up at equilibrium.

When ever a metal semiconductor junction is formed. we know that there are electrons in the conduction level of semiconductor which can move to the empty state above the Fermi level of the metal.

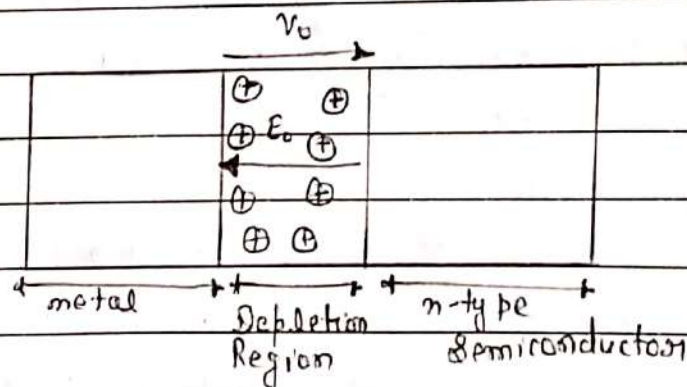
So, due to which a net negative charge will be there on the metal side and net positive charge on semiconductor side. which mean an electric field is setup. which will flow from semiconductor to metal side leading

to the contact Potential.

Now concentration of electrons in metal is of the order of 10^{22} cm^{-3} and electron density in semiconductor is 10^{17} cm^{-3} . So whenever a contact is formed between two, then due to low charge density on semiconductor side the electrons are not only removed from the surface but also from deeply inside the bulk of semiconductor.



(Before contact)



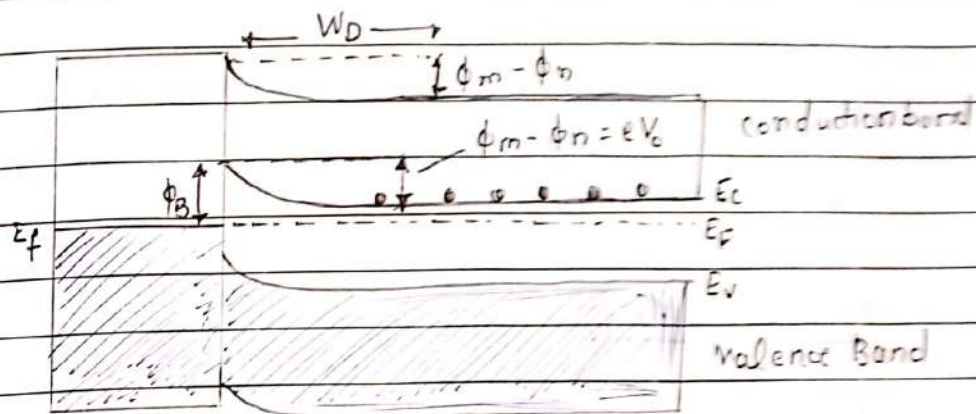
(Formation of contact Potential)

So there is a region within the semiconductor where electrons are lost as they move into the metal and this region is called the depletion region.

So now we have the m

electric field and contact potential (eV_0) between the metal and the semiconductor and this contact potential depends upon the difference b/w the work function ϕ_0 ,

$$eV_0 = \phi_{\text{metal}} - \phi_{\text{semi}} \quad \dots (1)$$



Now we know that the work function for the metal is a constant while in case of semiconductor it depends on the dopant concentration.

The contact potential then separates the barrier for the electrons to move from the n-type semiconductor to the metal. Initially, when the junction is formed electrons move to the metal to create depletion layer. The contact potential thus formed prevents further motion of the electrons to the metal. There is also barrier for electrons to move from metal to semiconductor. This is called the "Schottky barrier" and denoted by ϕ_B .

$$\phi_B = (\phi_m - \phi_n) + (E_c - E_{fn}) = \phi_m - \chi_n \quad \dots (2)$$

Where χ_n is electron affinity of n-type semi

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conductors at equilibrium, the flow of electrons from the semiconductor to metal is balanced by the contact potential so there is no net current.

By applying an external potential, the Schottky junction can be biased in two ways-

(1) Forward bias $\Rightarrow$

metal is connected to positive terminal and n-type semiconductor connected to negative terminal.

(2) Reverse bias $\Rightarrow$

metal is connected to negative terminal and n-type semiconductor connected to positive terminal.

The current flow depends on the type of bias and amount of applied external potential.

Forward bias $\Rightarrow$

In a forward biased Schottky junction, the metal is connected to the positive side and semiconductor is connected to the negative side of an external potential. We find that external potential is applied in such a way that it is opposite to the contact potential that is setup within the junction.

Since the region with the highest resistivity is the depletion region near the junction, the voltage drop takes place across the depletion region. As already mentioned above, under external bias, the Fermi level no longer line up but are shifted.

with respect to one another and the magnitude of the shift depends on the applied voltage.

Thus electrons injected from the external circuit into the n-type semiconductor have a lower barrier to overcome before reaching the metal. This leads to a net current in the circuit which further increases with the increase in external potential.

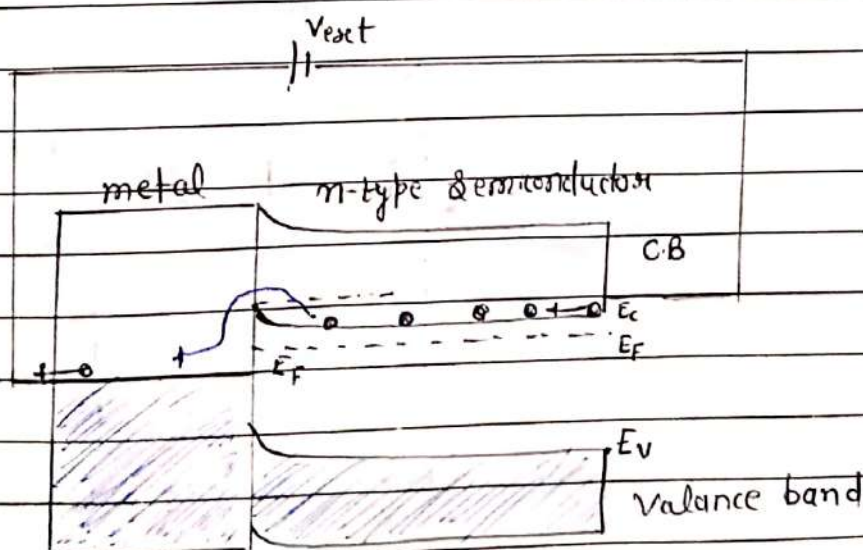
The current in a Schottky diode under forward bias -

$$J = J_0 \left[\exp\left(\frac{eV}{k_B T}\right) \right] \dots$$

where J is current density for applied potential of V .

J_0 is a constant and depends on the Schottky barrier (ϕ_B) for the system.

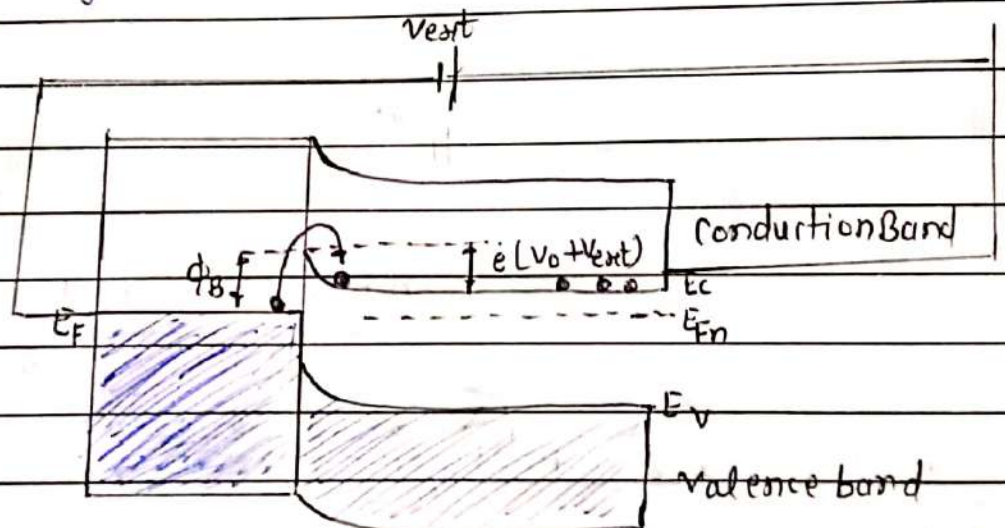
$$J_0 = \left[\exp\left(\frac{-\phi_B}{k_B T}\right) \right]$$



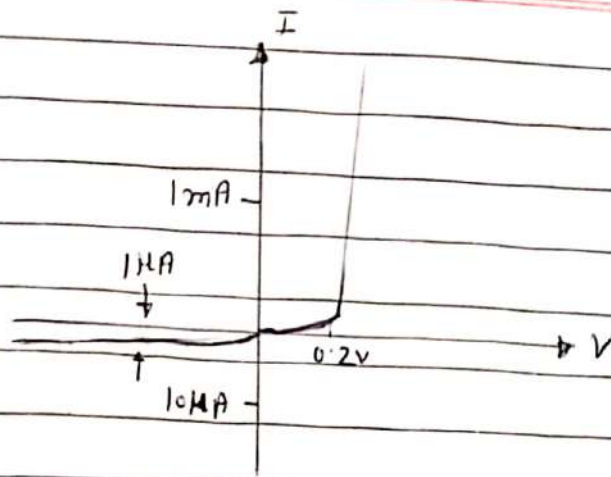
REVERSE BIAS :-

In case of a reverse bias, the metal side is connected to a negative and the semiconductor is connected to a positive. In this particular case, the external potential V_{ext} is in the same direction as the contact potential, which means the overall barrier goes up for electron motion from the n-type semiconductor to metal. There is a small current that goes from the metal to the semiconductor and this reverse current depends upon Schottky barrier (ϕ_B).

So there is a constant current in reverse whose magnitude is equal to J_0 . From calculations it shows that the current in the forward bias is order of magnitude higher than the current in the reverse bias. So Schottky junction act as a Rectifier.



Schottky Junction under reverse bias

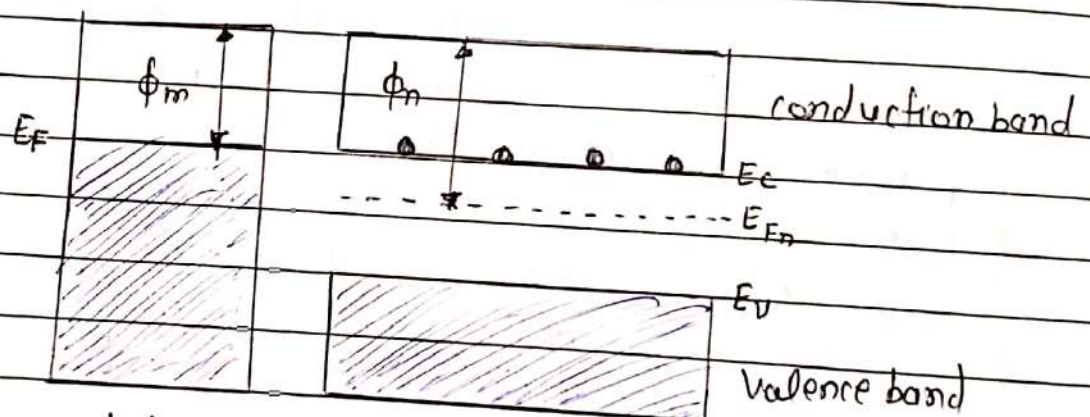


V-I characteristics of Schottky Junction showing Rectifying Property.

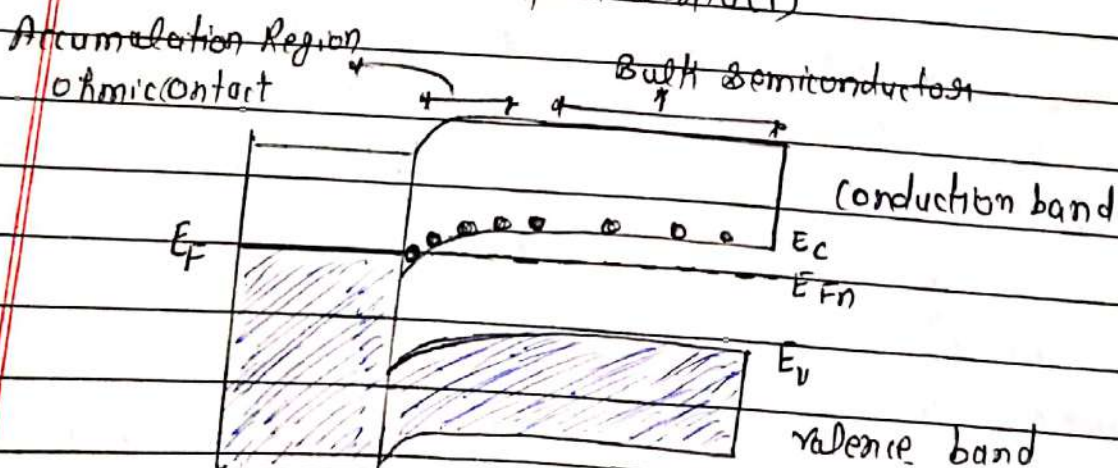
"Ohmic Junction"

A Schottky Junction is formed when the semiconductor has a lower work function than the metal, when the work function of the metal is smaller than the work function of the semiconductor, the junction formed is called "ohmic junction". The energy band diagram of the junction in equilibrium - There is a large number of electrons in the metal side and these electrons can move to the semiconductor side. So an electric field will again be setup and the field will be in the direction opposite to that of a Schottky junction. As electrons are going from the metal to the empty states in the conduction band, so, there is an accumulation region near the interface. The accumulation region has a higher conductivity than the bulk of the semiconductor due to this higher concentration of electrons. Thus, a ohmic junction behaves as a resistor conducting in both forward and reverse bias. The resistivity is determined by the bulk resistivity of the semiconductor.

One of the interesting application of ohmic junction is in thermoelectric devices. where a small volume can be cooled by application of direct currents. For ohmic junctions, depending on the direction of current flow, heat can be generated or absorbed. This is called "Peltier effect" and it is applicable to Junction between any two dissimilar materials. So when current flows through a metal semiconductor ohmic junction, heat will always be released or absorbed. This can be used as a practical cooling device. Both p and n type semiconductor are used and the current flow in such that one end is always cold while the other end is hot (heat released).



metal N-type Semiconductor
(before contact)

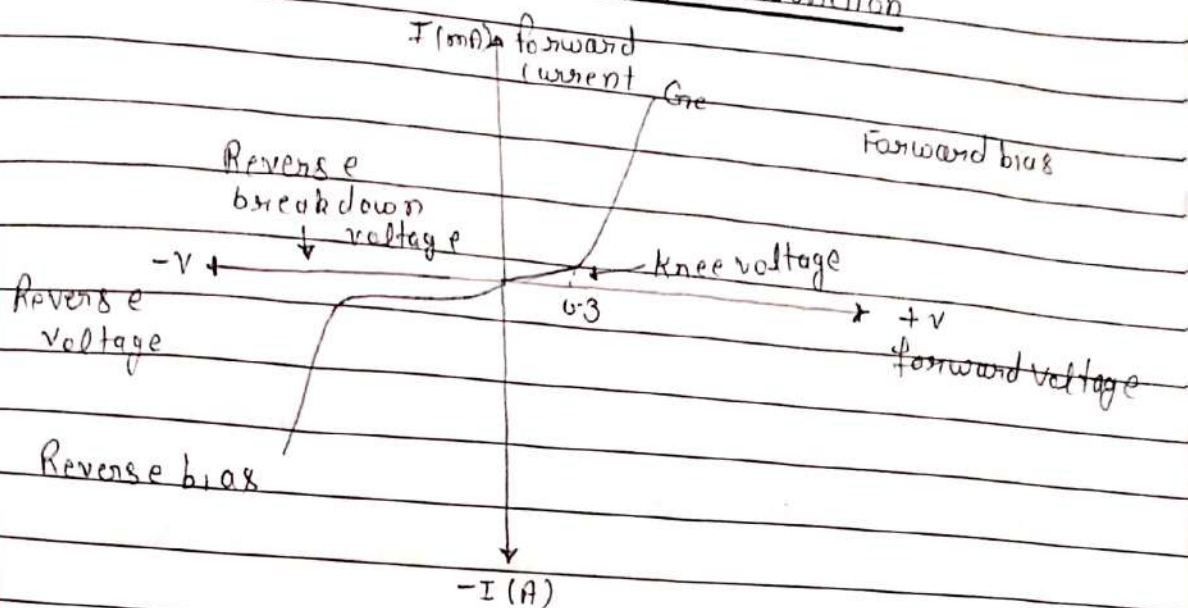


metal n-type semiconductor
(After contact)

Difference between Schottky and Ohmic Junctions

Schottky	Ohmic
1. Work function of metal is greater than work function of semiconductor.	1. Work function of Semiconductor is greater than metal.
2. Fermi level of Semiconductor is greater than Fermi level of metal.	2. Fermi level of metal is greater than Fermi level of Semiconductor.
3. Electric field will be set up from semiconductor to metal when contact is made.	3. Electric field will be set up from metal to Semiconductor when contact is made.
4. Energy level band up from Semiconductor to metal side.	4. Energy level band down from metal to semiconductor.
5. It can work as rectifier in Forward bias not reverse bias.	5. It can act as rectifier both in forward and reverse bias.

V-I characteristics of P-N Junction



- (1) In forward bias at forward voltage (0.3V for Ge and 0.7V for Si) the potential barrier across P-n Junction is eliminated and current starts flow. This voltage is called "Threshold voltage" or "Knee voltage".
- (2) As applied voltage further increases, the forward current rises exponentially as shown in graph.
- (3) In reverse bias, as the reverse voltage is increased from zero, the reverse current quickly rises to its maximum or saturation value. The slightly increase is due to impurities on the surface of semiconductor so current produced is called Surface Leakage current.
- (4) If reverse voltage is further increase Kinetic energy of minority carriers becomes so high that they knock out the electron from atom. At this stage junction breakdown occurs and there is sudden rise of reverse current.

"Carrier Transport - Drift and Diffusion"

There are two basic transport mechanism -

Drift $\Rightarrow$ movement of charge carriers due to electric field.

Diffusion $\Rightarrow$ flow of charges due to density gradient.

Drift Current $\Rightarrow$

When electric field applied to the semiconductor, it produces a force on electrons and holes so that they experience a net acceleration and net movement. This net movement of charge carriers due to electric field is called "Drift". The net drift of charge give rise to drift current. So, drift current density due to motion of electrons in conduction band is

$$J_e = n e \mu_e E$$

Where n = no. of free e^-

μ_e = mobility of e^-

E = electric field

Drift current due to motion of holes in valence band.

$$J_h = p e \mu_h E$$

So, total current density = $J = J_e + J_h$

$$J = n e \mu_e E + p e \mu_h E$$

$$J = e E (n \mu_e + p \mu_h)$$

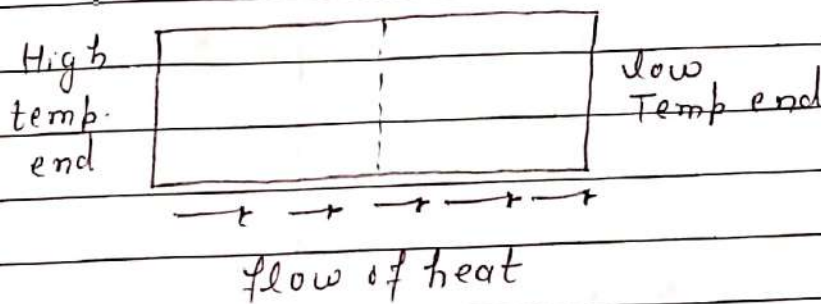
Note $\Rightarrow$

Drift current flows only when external electric field is applied across the semiconductor.

Diffusion Currents

In case of Semiconductors, current can also be flow without the application of external electric field.

Let us consider the material having two different temperatures across its two ends. Due to temperature gradient heat flows from higher temperature region to lower temperature region until the temperature is uniform throughout the material.



Similarly an electric current flow in Semiconductors even in the absence of applied electric field, provided a concentration gradient that exist in the material.

A concentration gradient exist if the number of electrons or holes is greater in one region of Semiconductor as compare to rest of region.

In Semiconductors charge carriers have tendency to move from the region of higher concentration to lower concentration. Thus movement of charges carriers takes place resulting in current called "Diffusion Current".

Diffusion current density due to flow of electrons

$$J_e (\text{diff}) = e D_e \frac{dn}{dx}$$

Diffusion current density due to flow of holes.

$$I_p(\text{diff}) = e D_p \frac{dp}{dx}$$

Difference b/w →

<u>Diffusion Current</u>	<u>Drift current.</u>
1. Diffusion current may occur even electric field not applied to semiconductor.	1. Drift current requires the presence of electric field.
2. Magnitude of diffusion current depends on the slope of carrier concentration.	2. Drift current depends on the concentration of semiconductor material.
3. Diffusion current flows in the direction of slope of carrier concentration and does not obey ohm's law.	3. Drift current flows in the direction of electric field and depends on ohm's law.

Avalanche Breakdown :-

The charge carriers of P-N Junction diode sometime absorb the heat from the environment at normal room temperature when reverse bias applied across the Junction. The kinetic energy of electrons increases and they start moving with high velocity. While moving they collide with other atom and Greater number of free electrons which causes reverse saturation current. Because of this saturation current the avalanche breakdown occur in diode.

Zener breakdown :-

Zener breakdown takes place when both sides of Junction are very heavily doped and consequently the depletion layer is thin and consequently the depletion layer is thin. When a small value of reverse bias voltage is applied a very strong electric field is set up across the

thin depletion layer. This electric field is enough to break the covalent bonds. Now extremely large number of free charge carriers are produced which constitute the zener current. Process is known as Zener break down.

Ques How does the doping increase the conductivity of Semiconductors?

Ans. The conductivity of semiconductors is increased by adding an amount of suitable impurity or doping. Doping can be done with an impurity which is electron rich or electron deficient as compared to intrinsic semiconductor, Si or Ge.

Such impurities introduce electronic defects in them. When silicon is doped with electron-rich impurities, the extra electron becomes delocalized. These delocalized electrons increase the conductivity of doped silicon due to the negatively charged electron, hence silicon doped with electron-rich impurity is called N-type semiconductor.

While electron deficit impurities increase the conductivity through positive holes and this type of semiconductor are called P-type semiconductor.

Difference between →

Avalanche Breakdown

1. Avalanche breakdown occurs because of collision of electrons.
2. It will destroy the Junction.
3. Pair of electrons and holes are produced
4. It occurs with low doping.
5. Breakdown voltage is directly proportional to temperature.
6. Temperature coefficient of avalanche breakdown is positive.
7. positive temp. coefficient means temperature of material increase with reverse voltage.

Zener Breakdown

1. Zener breakdown occurs because of high electric field
2. It will not destroy the Junction
3. Electrons are produced
4. It occurs with high doping.
5. Breakdown voltage is inversely proportional to temperature.
6. Temperature coefficient of zener breakdown is negative.
7. Negative temp coefficient means temperature decrease with reverse voltage.

Diode Equation :-

In PN Junction forward bias, beyond knee voltage, forward current rises exponentially according to following equation.

$$I = I_0 \left[\exp \left(\frac{V}{\eta V_T} \right) - 1 \right] \quad \text{--- (1)}$$

Where I = diode current

I_0 = diode reverse saturation current at room temp.

V = applied forward voltage

η = constant [1 for Ge + 2 for Si]

$V_T = \frac{kT}{q} = \frac{T}{11600}$ volt equivalent to temp

k = Boltzmann Constant = $1.38 \times 10^{-23} \text{ J/K}$

q = electronic charge = $1.6 \times 10^{-19} \text{ C}$

T = Temp (K)

In forward bias

$$\exp \left[\frac{V}{\eta V_T} \right] \gg 1$$

So equation (1), becomes

$$I = I_0 \exp \left[\frac{V}{\eta V_T} \right]$$

Reverse Bias :-

In Reverse bias, a ~~constant~~ constant reverse current saturation current I_0 flows due to minority carriers.

Here V is negative. So eq (1) can be modified as

$$I = I_0 \left[\exp \left(\frac{-V}{\eta V_T} \right) - 1 \right]$$

In Reverse Region,

$$\exp \left[\frac{-V}{\eta V_T} \right] \ll 1$$

$$\text{So } I = -I_0$$

Semiconductor materials for optoelectronic devices :-

Si and Ge are the commonly used semiconductors. They belong to group IV of periodic table and are used either in pure intrinsic or doped (extrinsic) form. The intrinsic and extrinsic semiconductors are known as "Elemental Semiconductor".

A compound semiconductor is a semiconductor composed of elements from two or more different groups of periodic table for Eg. one element from column III and one from column V, so called compound (III-V) semiconductor. Such as GaAs, InP, and GaN.

The elemental semiconductor Si and Ge are indirect band gap semiconductors and therefore not suitable for producing light.

So they are basically used for making rectifying diode while compound semiconductor are direct band gap semiconductors and have high optoelectronic conversion capacity, due to which they are widely used in fabrication of optoelectronic devices such as semiconductor lasers, LED, photodiodes, solar cell etc.

⇒ "Those device which is used to converting electronic energy to light energy called optoelectronic devices"

② Semiconductor materials for optoelectronic

Study of semiconductor materials began in the early 19th century.

(i) Elemental Semiconductor :-

The elemental semiconductors are those composed of single species of atom

Such as Si, Ge, Sn in column IV and Selenium (Se) and Tellurium (Te) in column VI of periodic table.

(2) Compound Semiconductors

There are numerous compound semiconductors, which are composed of two or more elements.

* GaAs (Gallium Arsenide) is a binary III-V compound which is a combination of gallium (Ga) from column III and Arsenic (As) from column V.

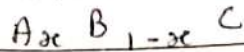
* Ternary Compounds can be formed by elements from three different columns e.g. Mercury Indium telluride (HgIn_2Te_4) a II III VI compound.

They can also be formed by elements from two columns such as aluminium gallium Arsenide ($\text{Al}_x\text{Ga}_{1-x}\text{As}$) which is ternary III-V compound where Al and Ga are from III and subscript x is related to composition of two elements from 100 percent Al ($x=1$) to 100 percent Ga ($x=0$).

Pure Si is most important material for integrated circuit applications and III-V binary and ternary compounds are most significant for light emission.

Fast diode laser :-

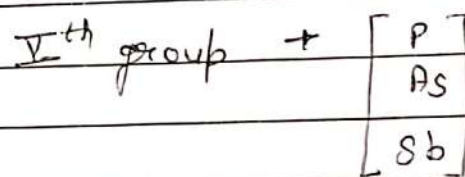
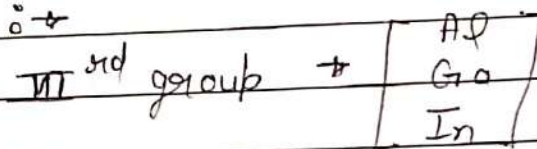
Ternary semiconductor from three and five columns of periodic table have general formula



Here A and B are from III column and C are from V column and vice versa.
 x = mole fraction.

3. (a) They are used element from III and V column are used in ternary semiconductor for making diode laser. That emit light in the wave length range of 0.7 μm to 1.0 μm . but not all ternary semiconductor are not good optic material

For Ex :-



In addition to III-V compounds some nitride compounds of aluminium nitride are used in making LEDs which emit light in the visible spectrum.

- (b) Gallium nitride (GaN) is used in semiconductor laser which emit blue beams. Gallium nitride Transistor are operate at high voltage and high temp. then GaAs, InN is used in infrared operation solar cell.

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(1) binary compounds IIIth and V as InSb (indium antimonide) is used in infrared detector and Thermal imaging sensor.

(2) II - VI $\rightarrow$ CdSe (Cadmium Selenide) are tested for high efficiency solar cell. Similarly CdS is used in photoresistor. It is also used in solar cell along with CdTe. Now CdTe is also used in photovoltaic cell.

Similarly these are compound like ZnSe, ZnTe from II and VI group. ZnSe is used in blue laser and LEDs, it is also a most common material for infrared optics.

ZnTe is used in making solar cell, blue laser and LED. when it is grown on AlSb, GaSb, PbSe etc.

(3) IV - VI $\rightarrow$ SnS₂ (Tin sulphide) it is used in gas sensing applications (Cu₂SnS₃) Copper Tin sulphide is used in Thin film solar applications and PbSnTe (Lead Tin Telluride) are used in IR detector and for Thermal imaging.

Questions

1. What do you mean by P-N Junction? How it is formed. Explain the working of P-N Junction diode and that is forward biased and Reverse biased? Explain its characteristics (V-I diagram).
2. Difference b/w diode ideal characteristics and diode Real characteristics?
3. Write down Metal Semiconductor Junction.
4. What is Semiconductor give its characteristics.
5. Difference b/w intrinsic and Extrinsic Semiconductors with Examples.
6. How doping (N-type, P-type) help in increasing the conductivity of semiconductor?
7. Explain the chemical bond in N-type and P-type.
8. Difference b/w ^{zener} break down and Avalanche breakdown.
9. Explain diode Equation.
10. What are optoelectronic devices explain in briefly the use of Semiconductor materials in optoelectronics?
11. Difference b/w Schottky and Ohmic Junction.
12. Draw and explain working of Schottky Junction in forward bias and Reverse bias.